# Unit – 3
# Mixed Reality
# Future Fictions Around the Principles of Interaction

# Future Fictions Around the Principles of Interaction

- It refers to imagining or designing speculative future scenarios that demonstrate how users might interact with digital and physical environments in advanced Mixed Reality systems.
- These “future fictions” help researchers and designers explore new interaction principles, technologies, and user experiences before they actually exist.

# Introduction

- Future fictions are **conceptual stories, scenarios, or prototypes** that depict how people might interact with MR technologies in the future.
- They are used to:
  - Predict **future interaction methods**
  - Explore **new MR interface designs**
  - Identify **challenges and opportunities**
  - Inspire **innovative MR applications**
- These ideas often combine **design thinking, speculative design, and human-computer interaction (HCI)** concepts.

# Principles of Interaction

1. **Natural Interaction**

- Users interact with digital objects using **natural gestures, voice, or body movement**.

- Example future fiction:
  - A student in an MR classroom **grabs a floating 3D molecule** and rotates it with their hands.

# Principles of Interaction

## 2. Context-Aware Interaction

- MR systems understand the **user's environment, location, and activity**.
- Example future fiction:
  - An MR system automatically shows **repair instructions** when a technician looks at a machine.

# Principles of Interaction

## 3. Spatial Interaction

- Users interact with digital objects **placed in real-world space**.
- Example future fiction:
  - Architects walk around a **full-scale holographic building model** in an empty field.

# Principles of Interaction

## 4. Collaborative Interaction

- Multiple users interact with the **same MR environment simultaneously**.
- Example future fiction:
  - Doctors from different countries collaborate on a **3D holographic human organ** during remote surgery planning.

# Principles of Interaction

**5. Multimodal Interaction**

- Future MR systems combine different input methods such as:
  - Gesture
  - Voice
  - Eye tracking
  - Touch
- Example future fiction:
  - A user selects a hologram **by looking at it and confirming with a voice command**.

# Examples:

1. Smart MR Classroom
2. MR Healthcare Training
3. MR Workplace Collaboration
4. MR Smart Cities

# Importance of Future Fictions in MR Research:

- Future fictions play a crucial role in MR development because they:
  - Encourage creative exploration of future technologies
  - Help designers understand potential user needs
  - Identify technical and ethical challenges
  - Guide the development of new MR interaction models

# Challenges:

- **Technical Challenges**
  - Limited processing power
  - Hardware constraints
  - Battery limitations
- **Usability Challenges**
  - Complex interaction design
  - User adaptation difficulties
- **Ethical and Privacy Issues**
  - Data privacy concerns
  - Security risks
  - Social acceptance of MR technology

# *Thank You…!!!*